AEGIS Digital Umbrella - Setup and Run Guide

This README is a beginner-friendly setup guide for a new Windows PC. AEGIS has a React frontend, a FastAPI Python backend, and a MongoDB database. See FEATURES.pdf for the full feature list. Follow the steps in order. The frontend and backend each need their own Command Prompt window and must both stay open while you use the app.

What the words mean

• **React frontend** is the website you see in the browser. `npm install` downloads its packages, and `npm start` opens it locally.

• **FastAPI backend** is the Python service that handles accounts, scans, reports, and database requests. It is started by Uvicorn with the command below. This project is **not Flask**; do not use `flask run`.

• **Virtual environment (`.venv`)** is a private Python package folder for this project. The commands below create it and install the backend packages into it.

• **MongoDB** stores accounts and scan history. Gemini and email are optional services for AI answers and password-reset email.

Step 1: Install the required programs

Use a web browser to install these on the new PC:

1. Install Python 3.14 for Windows (64-bit) from `https://www.python.org/downloads/windows/`. Keep the Python Launcher option enabled so the `py` command works.

2. Install Node.js 24 LTS from `https://nodejs.org/en/download/`. npm is included with Node.js.

3. Have a MongoDB database connection ready. MongoDB Atlas is a hosted option. In Atlas, create a database user, allow this PC's public IP under Network Access, and copy the application's connection string.

After installation, open **Command Prompt** (search for Command Prompt in the Windows Start menu; do not use PowerShell for these instructions). Check that Python and Node.js are available:

```
py -3.14 --version
node --version
npm --version
```

Each command should print a version. If one says it is not recognized, close Command Prompt, finish installing that program, then open Command Prompt again.

Step 2: Put the project folder somewhere easy to find

Extract the provided AEGIS Digital Umbrella folder. A simple location is:

```
C:\AEGIS Digital Umbrella
```

Your actual location may differ. To open Command Prompt in the correct folder without typing a long path:

1. Open the extracted project folder in File Explorer.

2. Open its `backend` folder.

3. Click the File Explorer address bar, type `cmd`, and press Enter. A Command Prompt opens already in the backend folder.

Step 3: Create the backend environment and install Python packages

In that backend Command Prompt, enter each line and press Enter:

```
py -3.14 -m venv .venv
.venv\Scripts\python.exe -m pip install --upgrade pip
.venv\Scripts\python.exe -m pip install -r requirements.txt
copy .env.example .env
notepad .env
```

The first command creates the .venv folder. The next two install the packages listed for this backend. The last two create your private settings file and open it in Notepad.

In Notepad, replace the MongoDB example on the MONGODB_URL= line with the connection string from your own MongoDB Atlas setup. The value usually looks like this, but use the full URI Atlas gives you:

```
MONGODB_URL=mongodb+srv://DATABASE_USER:URL_ENCODED_PASSWORD@YOUR_CLUSTER.mongodb.net/-retryWrites=true&w=majority
DB_NAME=aegis_digital_umbrella
```

Keep the MONGODB_URL= and DB_NAME= names as shown. Replace the example username, password, and cluster host. If the database password contains characters such as @, #, or /, URL-encode those characters before putting the password in the connection string. Save Notepad and close it. Do not share the .env file; it contains private settings.

The copied template already includes local frontend/backend addresses and optional settings. Add a real GEMINI_API_KEY only if AI recommendations and the assistant are needed. Add SMTP settings only if password-reset emails are needed. For Gmail, use a Google App Password, not the normal Gmail password. The scanner and PDF reports do not require Gemini or SMTP.

Step 4: Start the backend

In the same backend Command Prompt, run:

```
.venv\Scripts\python.exe -m uvicorn server:app --host 127.0.0.1 --port 8000
```

Leave this window open. This is the FastAPI backend process. To check that it started, open http://127.0.0.1:8000/api/ in a browser. A short AEGIS API message means it is running. If startup shows a MongoDB error, recheck the URI and Atlas Network Access setting.

Step 5: Install and start the React frontend

Open a **second Command Prompt**. In File Explorer, open the project's frontend folder, click the address bar, type cmd, and press Enter.

Run these commands in the second window:

```
copy .env.example .env
npm install
npm start
```

npm install is a one-time setup and may take a few minutes. npm start runs the React frontend. If asked whether to use a different port, type Y and press Enter, or close any other app using port 3000.

The app opens at http://localhost:3000. If it does not open by itself, type that address into your browser. Leave both Command Prompt windows open while using the app.

Step 6: Sign in and use the app

On a new empty database, choose Sign up to create an account, then sign in. To see existing accounts and scan history from another PC, configure the same MongoDB database used by the first PC. A different or empty database will not contain those existing records.

To run a scan, open Dashboard, choose New Scan, enter a website you own or are allowed to assess, select the checks, confirm authorization, and start. Open its result to review findings and recommendations. Scan History contains saved scans and PDF downloads.

Every later use

You do not need to reinstall packages each time. Start two Command Prompt windows in the appropriate folders.

In the backend folder, start the FastAPI service:

```
.venv\Scripts\python.exe -m uvicorn server:app --host 127.0.0.1 --port 8000
```

In the frontend folder, start the React website:

```
npm start
```

Keep both windows open. To stop either service, click its Command Prompt window and press Ctrl+C.

Common problems

• **`py` is not recognized:** Install Python and its Python Launcher, then close and reopen Command Prompt.

• **`py -3.14` cannot find Python:** Install Python 3.14, or use the matching installed Python version by changing `-3.14` in the virtual-environment creation command. After the `.venv` exists, use `.venv\Scripts\python.exe` for the other backend commands.

• **`npm` is not recognized:** Install Node.js from its official download page and reopen Command Prompt. These instructions use Command Prompt so PowerShell does not block the `npm.ps1` script.

• **MongoDB connection fails:** Check that the URI is complete, the username/password are for a MongoDB database user, special password characters are URL-encoded, and the PC's public IP is allowed in Atlas Network Access.

• **The site opens but cannot contact the API:** Make sure the backend window is still running. In `frontend\.env`, confirm `REACT_APP_BACKEND_URL=http://localhost:8000`, then stop and restart `npm start` after changing it.

• **`copy .env.example .env` says the destination exists:** The `.env` is already there. Keep it and edit it with `notepad .env`; do not overwrite it unless you mean to replace its settings.

• **AI recommendations are missing:** Confirm a valid `GEMINI_API_KEY` is saved in `backend\.env`, then restart the backend. AI is optional.

• **Password reset email does not arrive:** Confirm SMTP settings and use a Gmail App Password. Email is optional.

Authorized use

The built-in scanner performs bounded checks for SQL injection indicators, reflected XSS indicators, and CSRF token presence. It sends test inputs to the website. Scan only systems you own or have explicit permission to assess. Automated results need human review and do not prove a site is secure.